

EARLY-STAGE RESULTS

Emotion-Mediated Personalized Image Aesthetic Assessment

Predicting individual aesthetic preferences through emotional responses



What is Beauty?

AESTHETIC SUBJECTIVITY

The Subjectivity of Aesthetic Assessment

Beauty is not a single objective score. It is a multi-dimensional construct shaped by personal experiences, cognitive interpretations, and emotional resonance.

Emotional Resonance



Human judgment of beauty is closely bound to emotional reactions. A stunning mountain sunset evokes feelings of awe, calm, or nostalgia.

Cognitive Variations

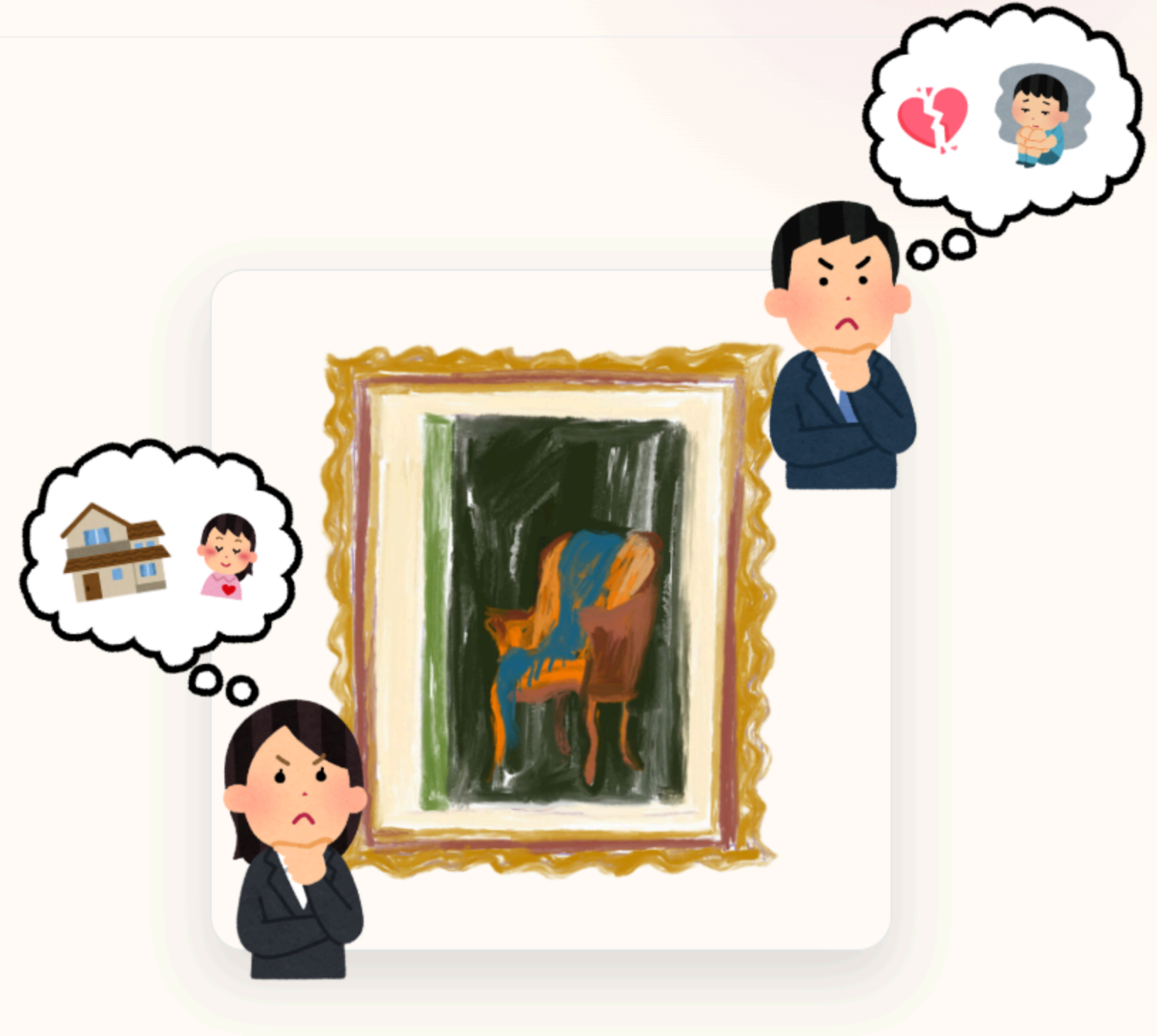


Individuals bring their unique cognitive backgrounds, personal experiences, and preferences to every viewing, leading to wide variations in ratings.

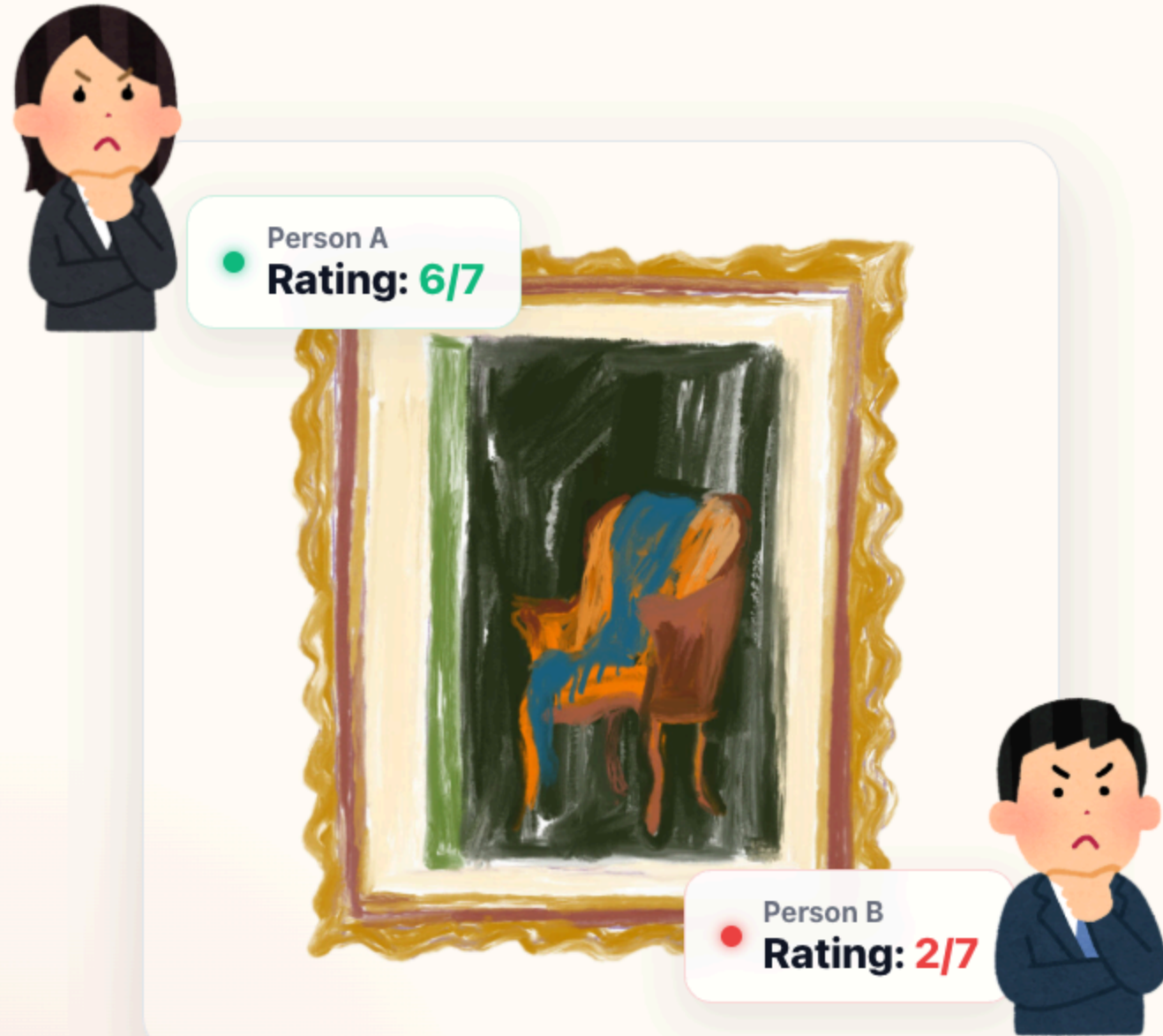
Beyond Numeric Regressions



Conventional models simplify beauty into a single number. Modern assessment requires capturing the emotional mediators behind that judgment.



The Core Problem.



The exact same image receives completely different beauty scores from different people.



Subjectivity of Beauty

Beauty is in the eye of the beholder. If we show the same picture to different observers, their responses exhibit high variance. Predicting a single average rating fails to capture individual preference.

What is PIAA?

Personalized Image Aesthetic Assessment (PIAA) is a specialized field in AI. Instead of predicting a generic general aesthetic score, the goal is to train an AI model capable of predicting the specific aesthetic score that a **particular individual** would assign to an image.

Our Ultimate Goal:

Predict the score for one specific person



Our Proposed Core Idea.

TRADITIONAL METHODS

Direct Prediction (Black Box)

Image → Deep AI → Beauty Score

Most previous methods map the image directly to a personal aesthetic score. It operates like a black box: we do not know why the model predicted that specific rating, and it lacks transparency.

OUR APPROACH

EXPLAINABLE

Emotion-Mediated Prediction

Image → Emotions → Formula → Personal Score

We introduce an explicit intermediate step: **predicted emotions**. Each user is modeled by a **personal formula** (emotion weights). The aesthetic score is computed as a weighted sum of these emotions, making the prediction fully interpretable.

📖 Grounded in Psychology Research

Our method is based on human psychological pathways: people do not judge beauty directly from raw image pixels. First, the image triggers an emotional response (e.g. nostalgic, amused, or impressed), and that feeling serves as the foundation for the aesthetic judgment.



Research Questions.

Q1 Accuracy vs. Direct Prediction

Can emotion-mediated modeling perform as accurately as direct aesthetic prediction?



Comparing our explainable two-stage pipeline against traditional black-box deep models to see if we sacrifice performance for transparency.

Q3 Psychological Grounding

Does the personal formula correlate with real-world personality traits?

Testing if users high in Openness to Experience show stronger weights for 'Intellectually Challenged' in their formula, confirming psychological grounding rather than random data fitting.

Q2 Cross-Category Consistency

Is a person's "personal emotion formula" consistent across different image categories?



If a user appreciates fine art due to nostalgia, do they use the same criteria for fashion or landscape photos, or does their formula shift?

Q4 Mediation Redundancy

Should 'Liked' or 'Beautiful' be excluded as mediating emotions?

Since these dimensions are close to the final aesthetic rating, does including them make prediction trivial? We report on both 9-emotion and 7-emotion models to ensure robustness.

The XPASS-Vis Dataset.

IMAGES

6,526

Art, Fashion, Landscape



PARTICIPANTS

129

Individual evaluators

TOTAL RATINGS

87,836

Fully labeled examples

LABELS PER RATING

1 + 9

1 Beauty (1-7) + 9 Emotions (1-5)



Note: Created by Hayashi-san's lab. The inclusion of granular, multi-dimensional emotional ratings per person is what makes our personalized approach uniquely possible. Most traditional datasets only contain general aesthetic averages.

List of 9 Rated Emotions

● POSITIVE

Liked it

Found it beautiful

Impressed

Motivated

Amused

● COGNITIVE

Nostalgic

Intellectually challenged

● NEGATIVE

Sad

Distasteful



The Prediction Pipeline.



Input Image

Any artwork, fashion photograph, or landscape scene.



TRAINED ON ALL USERS

Step 1: Shared AI

A generic model that predicts the 9 emotional response levels for the image (e.g. nostalgia = 4/5).



9 WEIGHTS PER USER

Step 2: Personal Formula

Each person has a unique set of weights representing how much they care about each emotion when judging beauty.



Personal Score

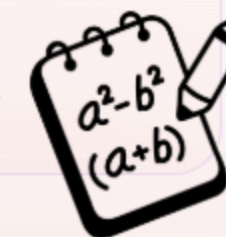
A custom aesthetic prediction tailored to that specific user.

THE PERSONAL FORMULA

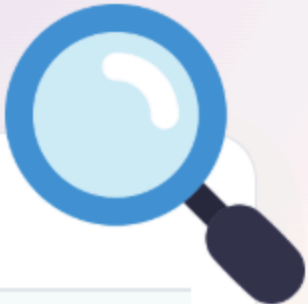


The formula simply evaluates: **How much does this specific person care about each of the 9 emotions when deciding if an image is beautiful?**

Because these weights are explicit, we avoid the black-box issue. We can look at the 9 numbers and read exactly what aesthetic priorities characterize each individual user.



Experiment 1: Is the Idea **Even Possible?**



METHOD / DATASET INPUT	ART	FASHION	LANDSCAPE	AVERAGE
All 9 emotions Using the full set of emotional responses	0.819	0.838	0.843	0.833
7 emotions (Real emotions) Removed circular proxies: 'liked' & 'beautiful'	0.716	0.705	0.740	0.720
Best previous method (ICI) Shi et al., 2024 (without emotions)	0.493	0.319	0.458	0.423

* Values represent Concordance Correlation Coefficient (CCC) score, max = 1.0 (higher is better).

Theoretical Ceiling: 0.720

Evaluating with Circular Proxies

Using all 9 emotions yields a prediction score of **0.833**. However, emotions like "liked it" and "found it beautiful" are circular references for aesthetic preference.

The Real Emotional Mediation

Removing circular proxies leaves 7 "pure" emotions. This still achieves a high score of **0.720**, nearly doubling the previous best method (**0.423**). This proves the feasibility of the concept.

Experiment 2: Where does Personalization Come From?



GAP 1 — FEELING DIFFERENTLY (0.39)

This represents observer subjectivity in emotional response. Different people look at the exact same image and feel completely different emotions (e.g. nostalgic vs neutral). This accounts for **88%** of the personalization drop.

GAP 2 — CARING DIFFERENTLY (0.05)

This represents observer difference in weightings. Even if two people feel the exact same emotions, they value them differently when assessing overall beauty. This accounts for only **12%** of the difference.

Crucial Insight: Personalization comes mainly from **how people feel about an image**, not how they weigh those feelings. This finding is completely unique to the XPASS-Vis dataset.

Experiment 3: Can We Predict Individual Emotions?



MODEL METHOD	SUPPORT = 10	SUPPORT = 25	SUPPORT = 50	SUPPORT = 100
Direct (No Emotion) Predicting aesthetic score directly from pixels	0.109	0.187	0.245	0.294
Through Emotion + Correction (Hybrid) Proposed emotion-mediated route	0.153	0.238	0.299	0.334
Improvement Relative performance gain	+40%	+27%	+22%	+14%

* Support represents the number of images rated by the person used to train their personalized formula.

ICI State-of-the-Art: 0.423 Theoretical Ceiling: 0.720

🚀 Highest Gain in Scarce Data

The improvement is largest (+40%) when we have the least data (only 10 images per user). This highlights how using emotional mediation acts as a strong prior, preventing overfitting.

📈 Consistent Outperformance

Going through predicted emotions and applying correction consistently beats direct score prediction at all support levels (10 to 100 images).

🔍 Room for Improvement

We are currently at 0.334, which is below ICI (0.423). However, the theoretical ceiling is 0.720, indicating that refining the emotion predictor offers a huge path forward.

Literature & Related Work.



PAPER	CONTRIBUTION	RELATION TO OUR WORK
ICI (Shi et al., 2024)	Predicts personal aesthetic score directly using image features and user metadata.	Ours uses emotions as an explainable intermediate step rather than a black-box end-to-end mapping.
HNEF (Lan et al., 2024)	Leverages emotional attributes as supplementary features to improve general (non-personal) aesthetic predictions.	Ours focuses on personalization, using emotional reactions as the primary signal to capture individual preference.
ligaya et al. (2021, Nature Human Behaviour)	Demonstrated that human aesthetic preferences can be modeled as a weighted linear combination of visual features.	Serves as the theoretical backing for our 'personal formula' concept (modeling beauty as a weighted sum of emotions).

🔖 **Key Takeaway:** Our approach is not just an arbitrary engineering heuristic. By grounding the "personal formula" in the weighted-feature preference model established by **ligaya et al. (2021)**, we align our AI architecture directly with how psychologists believe human aesthetic evaluation operates.

Challenges & Next Steps.

Research Challenges



Predicting Personal Emotions

An image can make different people feel very different things. A general AI model typically only learns the average human reaction, not individual emotional subjectivity.

Rare Emotion Sparsity

Certain emotions occur very rarely in particular domains, causing data imbalance that is hard to train.

Limited Personalized Data

Each participant rated ~100 images. Fitting the 9-dimensional personal formula on such limited samples poses a significant risk of overfitting.

Evaluation Alignment

To compare fairly against existing baselines, we need identical cross-validation data splits across all models.

Next Research Steps

01 Complete 5-Fold Cross-Validation

Run comprehensive experiments across all remaining folds to secure robust, stable benchmark metrics against ICI.

02 Correlate with Personality (Big Five)

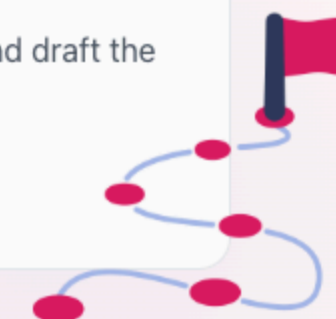
Cross-reference each person's emotion weightings with their Big Five personality profiles in XPASS-Vis.

03 Test Domain Consistency

Analyze whether an individual's personal formula stays consistent across different image domains.

04 Paper Write-up & Submission

Synthesize findings, document the explainability advantages of emotion mediation, and draft the paper.



Thank You!

Happy to take any questions or suggestions.